

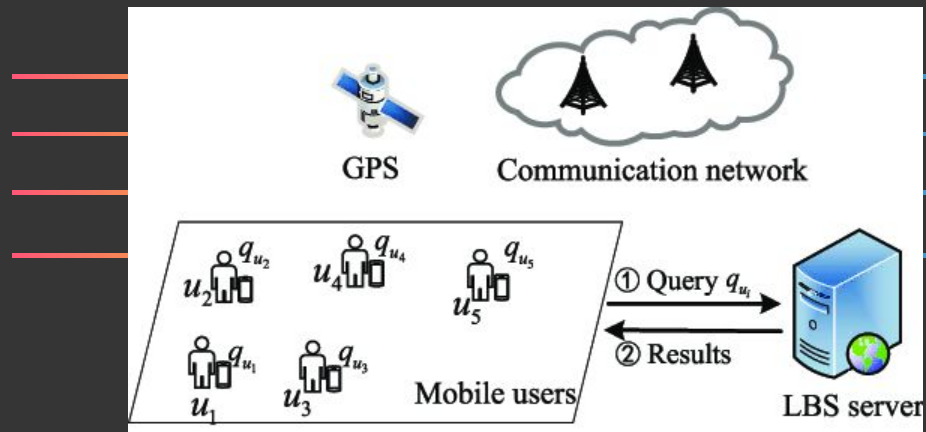
Approximate location a compromise between privacy and utility

Content

1.
System & Adversary model
2.
Scientific papers
3.
Creation of ApproxLoc
4.
Findings

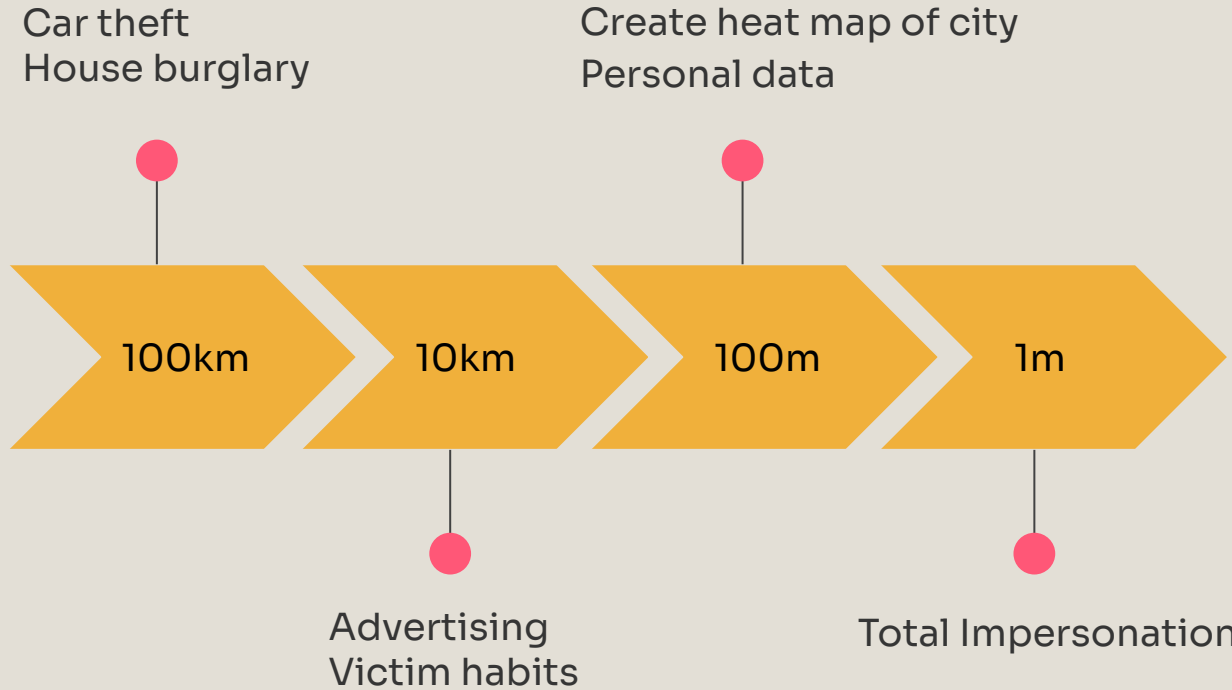
System

- GPS System
- Location Calculation (user)
- Sending Location through network
- LBS Server running
- Sending back the service



- GPS & Network safe
- LBS server compromised

Attacks on different granularity



State-Of-The-Art literature

Approximate location arrives in smartphones in 2020's (Android 12 / IOS 14)

<https://insidegnss.com/location-privacy-challenges-and-solutions/>

<https://dl.acm.org/doi/epdf/10.1145/3474717.3484219> (Introduction of metrics)

<https://doi.org/10.1080/17489725.2024.2310006> (UX)



Related Work 1

Which LBS is used ?

36 iphones + 36 android (in Germany) => evaluate around 30 apps (Bolt, Doctolib, Uber ...)

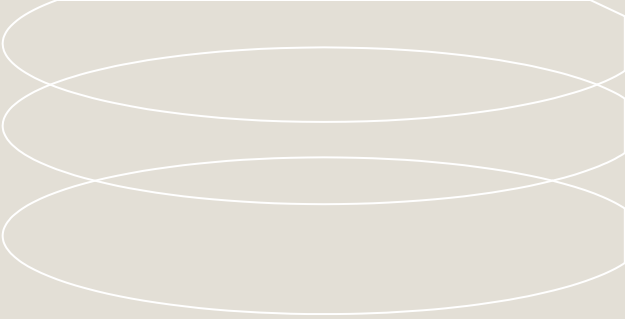
Conduct of study

Allow fine location on app to understand how it is supposed to work; switch to approximate location to view changes in the app.
View dark patterns, change in UI/UX, etc..

Utility vs privacy

graphs difference between ios & android

doesn't do a correlation between privacy and coarse location

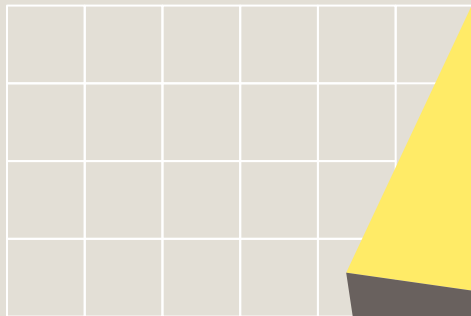


Sven Heitmann, Alexia Pagotto & Christian Kray (2025) Approximate vs. precise location in popular location-based services, Journal of Location Based Services, 19:1, 1-42, DOI: [10.1080/17489725.2024.2310006](https://doi.org/10.1080/17489725.2024.2310006)



Related Work 2

Liyue Fan and Ishan Gote. 2021. A Closer Look: Evaluating Location Privacy Empirically. In Proceedings of the 29th International Conference on Advances in Geographic Information Systems (SIGSPATIAL '21). Association for Computing Machinery, New York, NY, USA, 488-499.
<https://doi.org/10.1145/3474717.3484219>



Which LBS is used ?

RioBuses and geoLife for Data (from point A to point B)

Conduct of study

Developed a java program that process data and create metrics, especially “utility” (which is a set of number) for SpotMe,Cloaking,Noise techniques

Utility vs privacy

Utility is a set of measure

Privacy not well-defined -> if there is noise... there is already some privacy

Utility/Params
Hamming
Haversine (in m)
Tot Dist (in %)
Max Dist (in %)
Std Dev Displacement (in %)
Max Dist Home (in %)
Rad Gyration (in %)
Diff Places (in %)
Significant Places (in %)
Avg Mobility Error (in %)

ApproxLoc

WebApp accessible at :
r4ph3uz.github.io/EP2510



ApproxLoc stack

React



Leaflet



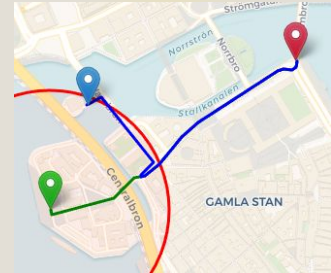
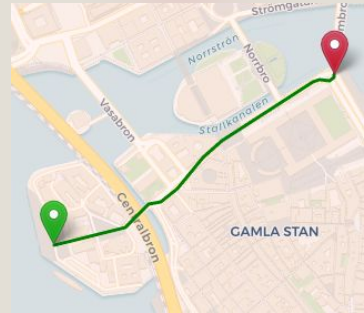
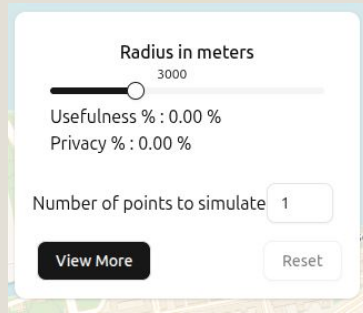
ShadCN



OpenStreetMap



WebApp



Equations

Privacy

privacy increasing with distance: as $d(a,b)$ grows, privacy approaches 100%

The radius parameter controls how fast privacy increases with distance (higher radius → faster increase)

$$1 - \exp(-\text{radius} * d(A,B))$$

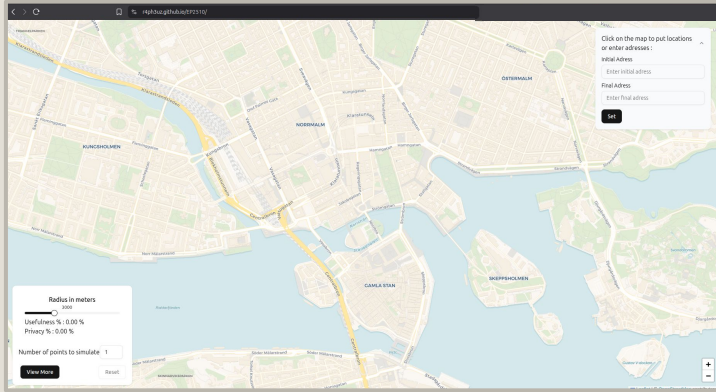
Usefulness

Usefulness measures how much of the path is shared: it is the ratio between the shared path length from A to B and C to B, on the total path length from A to C

A higher value means more overlap, hence more useful shared information.

$$\frac{\text{length_shared_path}(A,B,C)}{\text{length_total_path}(A,C)}$$

3 situations tested



Town

A lot of roads, where radius have high impact on usefulness and privacy

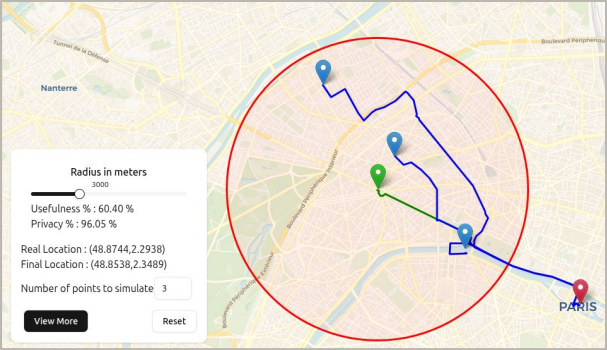
Suburbs

Often few roads with a main path, hence deviating from it implies less usefulness

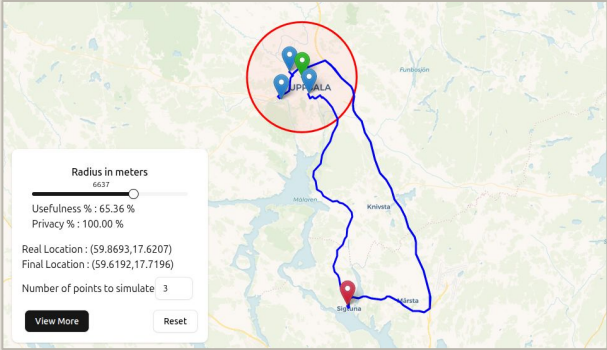
Countryside

Only one path, all points will go through a part of the path at one or another moment

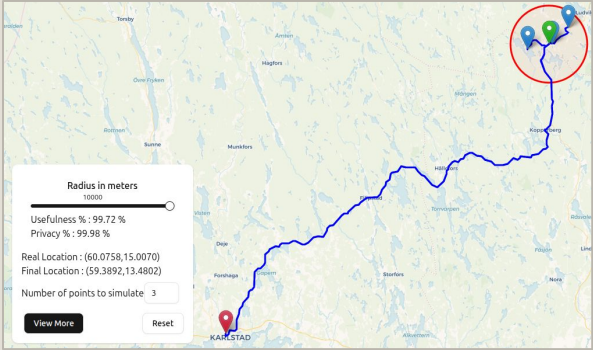
Town



Suburbs

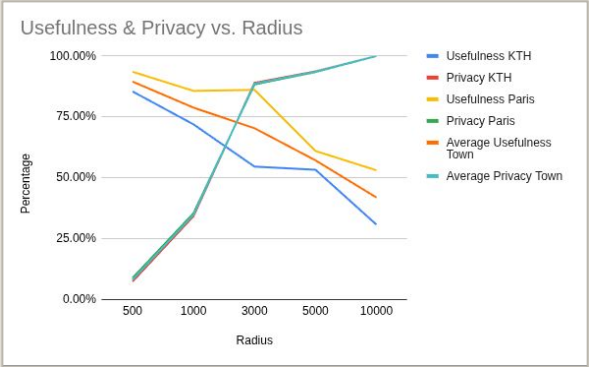


Countryside

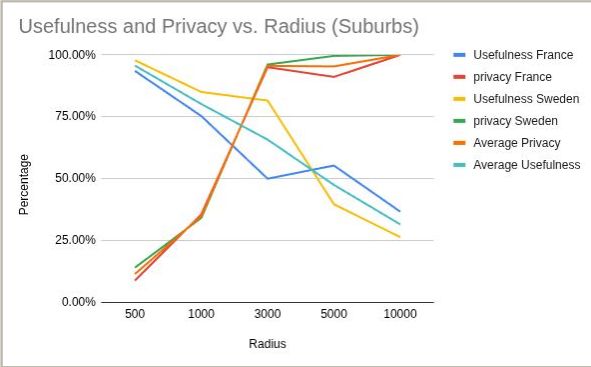


3 situations

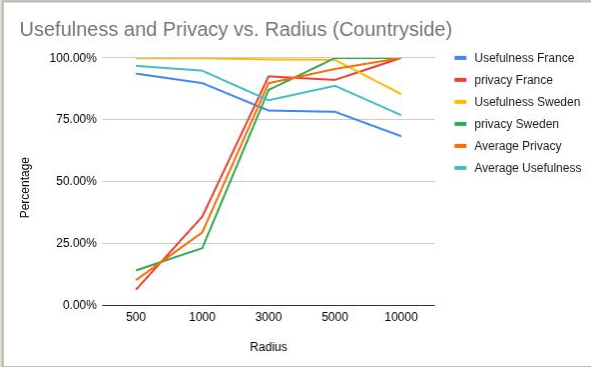
Town



Suburbs



Countryside



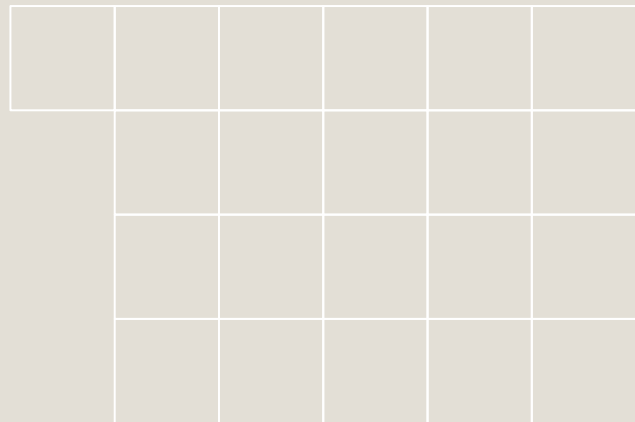
Thank you

Limitations and future work:

- find new formulas to suit reality
- improve UX on ApproxLoc
- France / Sweden
- implement other algorithms of Approximate Location

Strengths:

- open-source
- reproducibility





Thank you